

# Getting Started with Data Science and Data Analytics.

Introduction to Data Science, & Data Analytics is a comprehensive beginner's guide to analysis. This course covers a wide range of topics, including data ecosystems, data lifecycle, and the distinction between data science, & data analytics. Through case studies and real-world examples, learners will explore the applications of data analytics and data science in business. The course emphasizes the importance of data literacy and provides guidance on building your data and analytical skill set. Whether you're a business professional or an aspiring data analyst, this course is designed to empower you with the necessary skills and understanding to excel in the dynamic world of data science and analytics.

|                                      |  |                                                                                                                                                                                             |
|--------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisites for this course</b> |  | No prior knowledge or experience in data science, data analytics or business analytics is required. The course is designed for beginners who are interested in learning about these fields. |
|                                      |  |                                                                                                                                                                                             |
|                                      |  |                                                                                                                                                                                             |

## Course Breakdown:

### Part 1: Introduction and Overview

#### Session 1: Welcome and Course Objectives

- What is Data and How is it Generated in the Real World?
- Brief on the Importance of Data in Today's World
- Demand for Data-literate Professionals

#### Session 2: Data Science vs Data Analytics: What's the Difference?

- Five Main Evaluation Techniques: Scope, Objectives, Techniques, Applications, and Skillset

#### Session 3: Applications of Data Science & Data Analytics

- Industry-Specific Applications
  - Case Study 3: Social Media Analytics
    - How Data Analytics tools are used to gauge customer sentiment and tailor marketing strategies.



## **Part 2: Building Your Data & Analytical Skill Set**

### **Session 1: Core Competencies in Data & Analytics**

#### **❖ Data Literacy**

- Case Study: Data-Driven Decision Making in E-commerce
  - How an online retailer used data literacy to interpret customer behavior metrics and improve sales.

#### **• Critical Thinking Skills**

- Example: Crisis Management in Airlines
  - How critical thinking was employed to manage and mitigate a PR crisis using real-time data.

#### **❖ Hypothesis Formation and Testing**

- Case Study: A/B Testing in Digital Marketing
  - How hypotheses are formed and tested to optimize website conversions.

#### **❖ Data Wrangling (Data Cleaning & Transformation)**

- Example: Data Preprocessing in Healthcare
  - The role of data wrangling in cleaning electronic health records for better predictive models.

#### **❖ The Ability to Visualize Data**

- Case Study: Visual Analytics in Sports
  - How data visualization is used to analyze player performance in professional sports.

#### **❖ Basic to Advanced Knowledge in Mathematics**

- Example: Algorithmic Trading
  - The role of mathematical models in automated trading strategies.

#### **• Data Science and Data Analytics Overlapping Skills**

- Case Study: Customer Churn Prediction
  - How both Data Science and Data Analytics skills are used to predict and prevent customer churn.



## Session 2: How to Improve Your Skills?

### ❖ Embrace the Challenge

- Confronting mental barriers
- Building confidence in data analysis
- Cultivating a growth mindset

### ❖ Considering Opposing Viewpoints

- Expanding perspectives
- Challenging biases and assumptions
- Practicing critical thinking and analysis

### ❖ Learning from Real-World Examples

- **Case Study: Fraud Detection in Banking**
  - Studying successful data analysis cases and defining the purpose and objectives of analysis.
  - Examining industry-specific use cases & identifying relevant questions and hypotheses.
  - Exploring different analytical approaches and tools & applying lessons learned to your analyses.

### ❖ Continuous Learning and Improvement

- Staying updated with industry trends and advancements
- Taking advantage of online courses and resources
- Practicing regularly and seeking feedback